

EMBU, Kenya

WMO: 637200

Lat: 0.5007S

Lon: 37.4591E

Elev: 4900

StdP: 12.27

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	55.3	56.8	43.1	49.3	69.0	45.9	54.8	67.9	10.5	65.0	9.3	66.1	2.0	0	0.264

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.7	83.8	63.3	82.2	63.2	80.7	63.3	67.2	76.5	66.5	75.8	65.9	75.2	8.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	108.4	71.5	63.4	105.0	70.7	62.8	102.9	70.1	34.7	76.7	34.1	76.2	33.7	75.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.1	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			48.0	69.1	2.4	1.3	46.3	70.1	44.9	70.8	43.6	71.6	41.9	72.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.0	69.0	70.7	71.0	69.8	68.4	65.7	64.3	65.1	67.6	69.3	68.2	67.7		
	DBStd	3.00	1.71	2.07	2.63	2.11	1.57	2.17	2.47	2.60	2.68	2.29	1.73	1.58		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	108	0	0	0	0	1	18	45	32	9	1	1	1		
	CDD50	6586	589	581	652	595	572	470	442	467	526	597	545	550		
	CDD65	1220	125	161	187	145	107	37	22	34	86	133	97	86		
	CDH74	4676	647	1044	1157	402	134	25	35	73	340	461	166	192		
	CDH80	471	32	151	210	32	1	0	0	1	19	24	1	0		
(14) Wind	WSAvg	3.6	5.7	6.1	4.5	3.3	2.4	2.0	2.1	2.4	3.1	3.5	3.7	4.5		
(15) Precipitation	PrecAvg	46.2	1.2	1.1	4.2	11.1	6.6	1.1	1.1	1.4	1.2	6.4	9.2	2.1		
	PrecMax	75.9	5.7	6.1	8.8	22.2	27.1	5.2	7.1	4.1	3.6	24.4	23.8	8.6		
	PrecMin	25.1	0.0	0.0	0.0	2.5	0.6	0.0	0.0	0.0	0.0	0.0	2.2	0.0		
	PrecStd	13.2	1.6	1.6	2.6	4.6	4.6	1.0	1.1	0.9	0.9	5.2	4.7	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.8	85.5	86.1	83.2	78.8	76.3	77.1	78.8	82.3	82.5	79.2	79.3		
		MCWB	62.6	62.6	63.9	64.2	65.5	63.6	61.8	61.8	62.9	64.1	65.1	63.9		
	2%	DB	80.8	83.3	84.2	80.7	76.9	74.4	74.6	75.7	79.7	80.5	77.3	77.7		
		MCWB	63.1	62.5	63.6	64.3	65.2	62.8	61.2	61.2	61.8	63.8	64.9	63.9		
	5%	DB	79.2	81.6	82.6	78.2	75.4	73.1	72.6	73.7	77.8	78.6	75.8	76.3		
		MCWB	63.2	62.8	63.3	64.7	64.5	62.3	60.8	60.7	61.1	63.2	64.7	63.9		
	10%	DB	77.7	80.0	80.7	76.3	74.1	71.5	70.3	71.6	75.6	76.7	74.4	75.0		
		MCWB	63.2	63.0	63.4	64.8	64.2	62.0	60.2	60.2	61.0	63.0	64.5	63.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.1	66.8	67.5	68.0	67.2	65.4	63.7	63.9	66.2	66.7	67.2	67.5		
		MCDB	77.1	79.0	78.7	76.9	75.5	73.1	73.6	74.9	78.7	77.4	75.8	75.4		
	2%	WB	65.9	65.6	66.4	66.9	66.2	64.3	62.3	62.5	63.5	65.5	66.3	66.5		
		MCDB	75.9	78.2	78.1	75.4	74.4	72.1	71.4	72.7	75.5	75.9	74.5	74.6		
	5%	WB	65.3	64.7	65.7	66.2	65.4	63.2	61.5	61.5	62.4	64.7	65.6	65.6		
		MCDB	75.5	77.5	77.2	74.4	73.4	70.6	70.0	71.0	74.2	74.9	73.7	73.7		
	10%	WB	64.5	63.9	64.9	65.4	64.7	62.5	60.7	60.6	61.6	64.0	64.9	64.7		
		MCDB	74.9	76.7	76.2	73.5	72.4	69.6	68.6	69.5	72.6	73.5	72.9	72.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	21.3	19.7	14.9	13.0	12.1	12.6	13.6	17.1	16.4	13.8	16.0		
		MCDBR	22.0	23.0	23.0	17.1	14.6	14.7	16.3	17.1	20.8	19.0	15.9	18.1		
	5% WB	MCWBR	10.8	9.3	8.7	6.2	5.7	6.0	6.3	6.3	7.2	6.4	6.9	9.1		
		MCWBR	18.5	20.6	18.6	14.9	13.6	13.0	14.2	15.1	18.4	16.3	14.5	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.367	0.387	0.381	0.341	0.336	0.389	0.402	0.376	0.354	0.355	0.356	0.366		
	taud		2.464	2.388	2.435	2.599	2.606	2.406	2.343	2.414	2.476	2.501	2.522	2.485		
	Ebn at Noon		305	299	299	304	298	278	276	289	302	306	307	304		
	Edh at Noon		37	40	38	31	30	36	39	37	36	36	35	36		
(44) All-Sky Solar Radiation	RadAvg		1978	2081	2010	1793	1695	1460	1352	1441	1796	1843	1714	1804		
	RadStd		145	105	148	103	59	105	139	98	109	79	91	118		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page